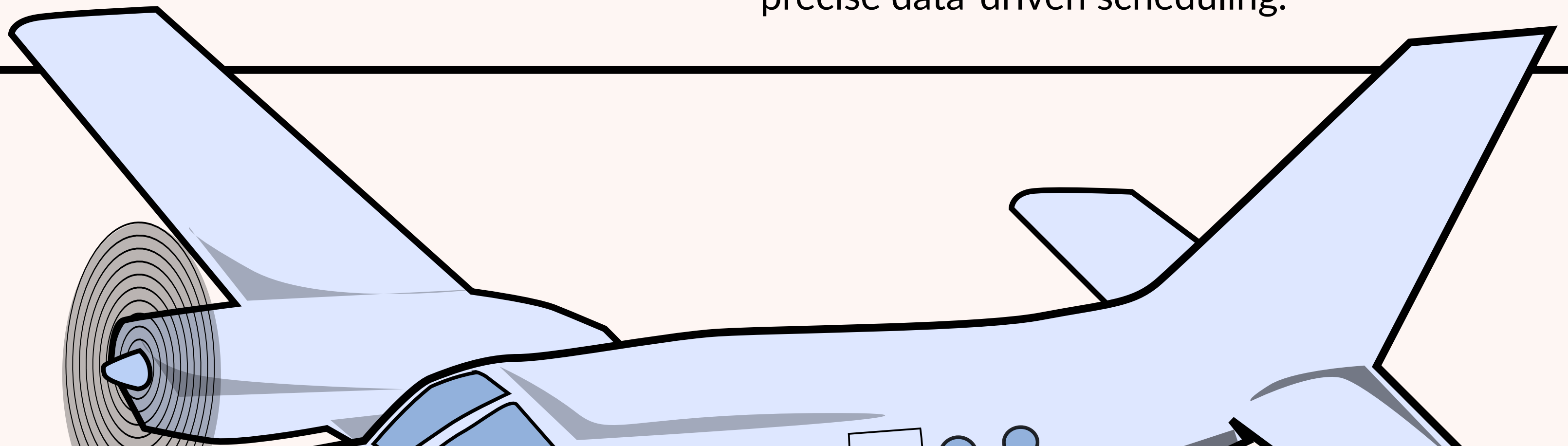
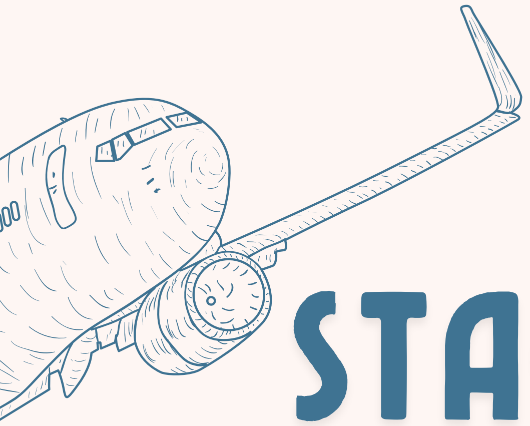
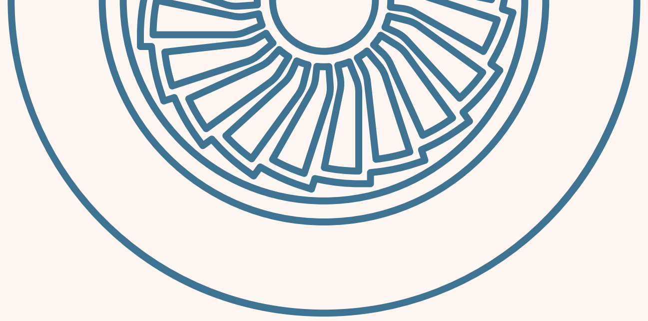
STAGE 1: EMPATHIZE – 5W1H FRAMEWORK

Who will view this Dashboard?	What problem does this dashboard solve?	When and where will stakeholders view this dashboard?	Why do stakeholders need this dashboard?	How have stakeholders tried to achieve this goal?
Senior Aviation Leadership, Hub Managers, and Data Analysts.	It synthesizes fragmented flight logs and delay codes to link operational delays directly with specific airlines and weather events.	**When:** During daily operations meetings and when severe weather hits. **Where:** Operations Control Center (OCC)	To quickly grasp the root causes of delays without manual data mapping and to execute early-stops/cancellations on high-risk routes.	Doing manual data matching in Excel to cross-reference FAA flight logs with internal schedules, which is slow and prone to errors.
If only one key stakeholder is chosen, who would it be?	Senior Aviation Leadership & Hub Managers.			
Summarize the problem in one sentence	This dashboard helps stakeholders unify national flight logs with delay codes to identify which airlines and hubs generate the most delays, enabling precise data-driven scheduling.			

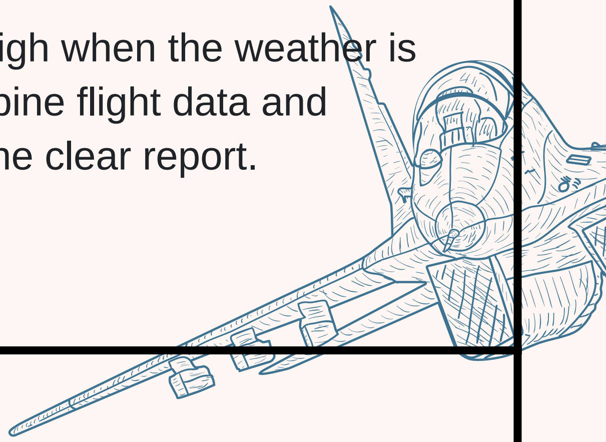




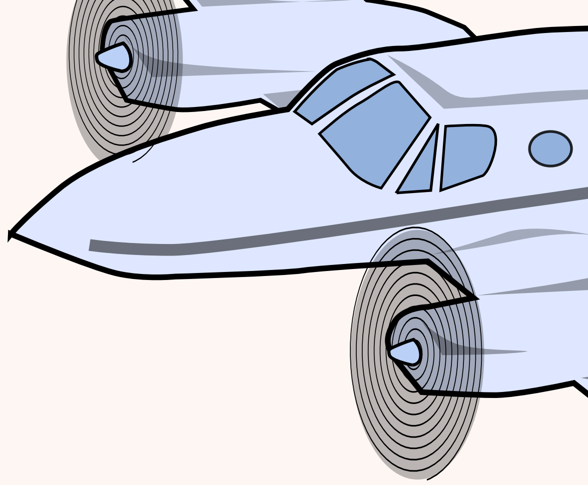


STAGE 1: EMPATHIZE MAP

What does the stakeholder think and feel?	What does the stakeholder see?	What does the stakeholder say and do?
• **Thinking:** Doubts whether "weather" is truly the primary cause of all delays. • **Feeling:** Frustrated by the disconnect between internal reports and the FAA's total delay numbers, and confused by unstable OTP margins day over day.	Fragmented reports: Operations team shows weather delays; FAA shows total delays. Unclear connection between morning minor delays and evening total network collapses.	"Why are our delays so high when the weather is clear?" Asks staff to combine flight data and cancellation codes into one clear report.
What are the biggest problems and challenges?	**What are the opportunities and benefits?**	
No clear overview of "Late Aircraft" cascading effects. High on-time rates in the morning conceal poor actual reliability in the evening.	Quickly identify and halt unprofitable schedules on severely delayed routes. Maximize predictability in On-Time Performance (OTP).	



STAGE 2: STAKEHOLDERS JOURNEY



Step 1

Starting from the company's operational situation, Leadership notices that despite allocating ground turnarounds, delays are rampant. They want to understand the true efficiency of the national network.

Step 2

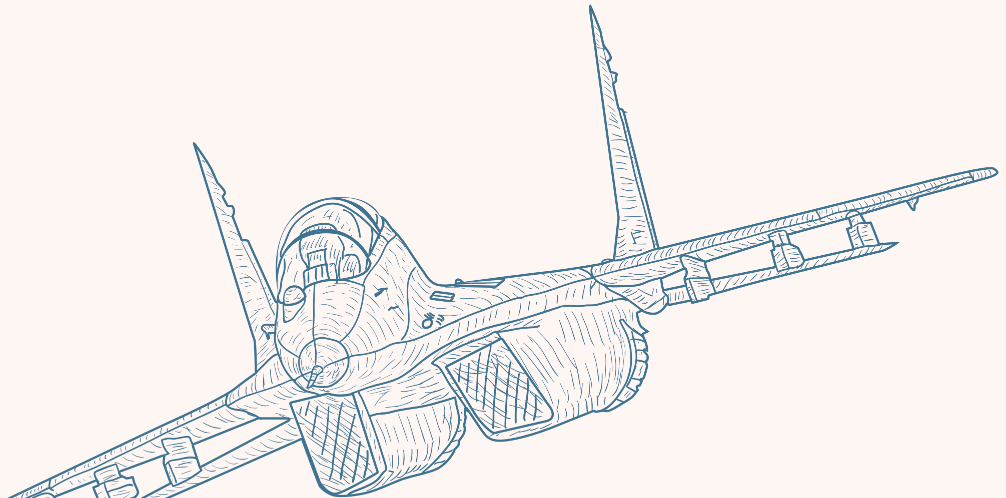
To get the needed insights, stakeholders reach out to Data Analysts to build dashboards that link flight logs directly with delay causes and airport geography.

Step 3

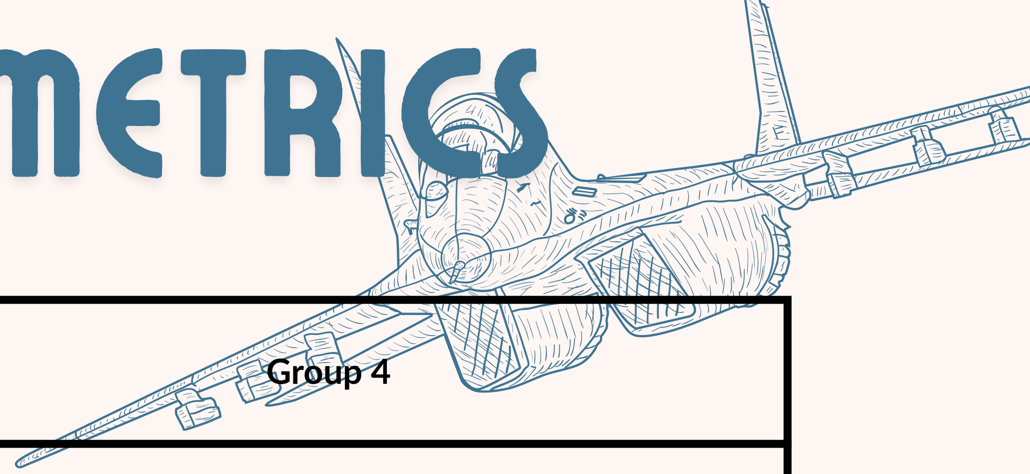
After receiving the dashboard, stakeholders find clear answers: tracking delays by hour, exposing airlines "burning fuel" via scatter plots, and comparing controllable vs uncontrollable delays. They group to brainstorm schedule reallocations.

Step 4

New automated scheduling rules are launched (e.g., proactive cancellations for Q1 winter storms based on KPIs), ground buffer times are widened, and performance data is monitored via 4 designated macro tracking pages.



STAGE 3: DEFINE POU & NORTHSTAR METRICS



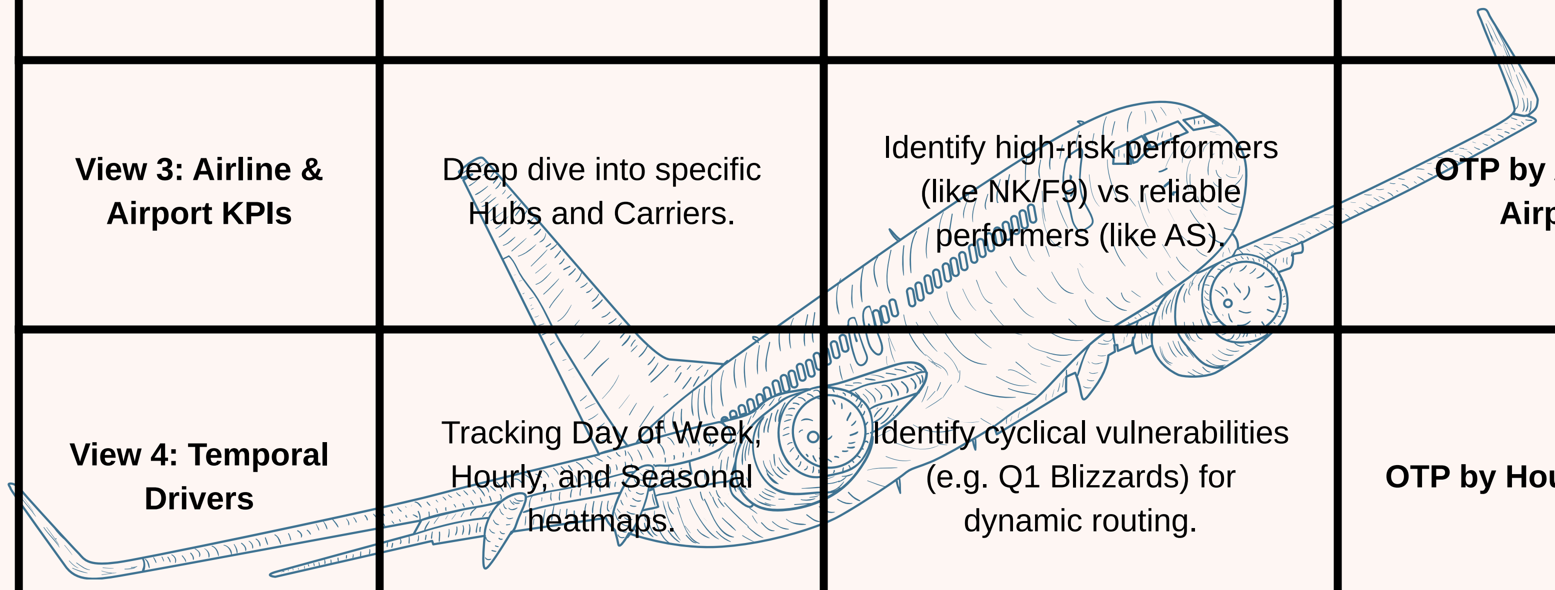
Dimension Data Group:

Group 1	Group 2	Group 3	Group 4
Geography / Routing (Origin Airport, Destination Airport, City, State)	**Carrier Identity** (Airline Name, IATA Code)	**Time** Year, Month, Week, Day, Hour	**Delay Reasons** (Weather, Late Aircraft, NAS, Security, Airline)

NORTHSTAR 1	NORTHSTAR 2 (Optional)
What VALUE you want to measure? Operational Punctuality	**What VALUE you want to measure?** Operational Severity / Waste
WHEN the value DELIVERY SUCCESS? When the On-Time Performance (OTP) grows steadily, overcoming erratic fluctuations across days/weeks.	**WHEN the value DELIVERY SUCCESS?** When severe delays (>15 mins) and cascading 'Late Aircraft' time significantly drop.
Northstar Metric Name On-Time Rate (OTP)	**Northstar Metric Name** Total Delay Minutes
WHY do you choose this metric? High volume of flights doesn't mean much if they are constantly delayed. OTP is the ultimate bottom-line goal to ensure reliability.	Directly measures the severity of the bottleneck. Helps identify which delays are minor vs burning massive fuel (cost).

Views & Breakdown Formulas:

View	Description	Why	NORTHSTAR Formula	NORTHSTAR 1 (Profit Margin)	NORTHSTAR 2 (ROAS)
View 1: Operations KPIs	High-level macro view of OTP, Volume, and total delays.	Assess overall operational health of the national network.	**OTP = Total On-Time Flights / Total Flights**	Avg Delay = Total Delay Minutes / Total Delayed Flights	View 1: Operations KPIs
View 2: Delay Root Causes	Granular metrics on Controllable vs Uncontrollable delays.	Pinpoint exact factors (e.g., Late Aircraft) to execute proactive schedule stops.	**% Controllable = (Airline + LateAircraft) / Total Delays**	Total Delay Minutes by Cause	View 2: Delay Root Causes
View 3: Airline & Airport KPIs	Deep dive into specific Hubs and Carriers.	Identify high-risk performers (like NK/F9) vs reliable performers (like AS).	**OTP by Airline / Airport**	Avg Delay by Route / Airport	View 3: Airline & Airport KPIs
View 4: Temporal Drivers	Tracking Day of Week, Hourly, and Seasonal heatmaps.	Identify cyclical vulnerabilities (e.g. Q1 Blizzards) for dynamic routing.	**OTP by Hour / Season**	Total Delay Minutes by Month	View 4: Temporal Drivers



STAGE 3: IDEATE – BRAINSTORMING

Start With: Based on the NorthStar Metric, brainstorm breakdown dimensions and relevant insights from different perspectives.

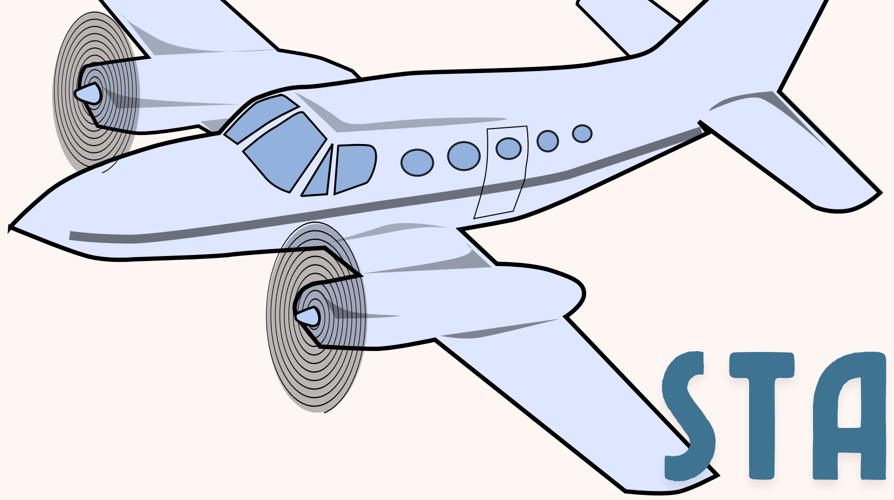
Overview Layer	**Metric 1** On-Time Performance Rate (OTP)	**Metric 2** Total Flights	**Metric 3** Total Delay Minutes	**Metric 4** Cancellation Rate (%)
	Metric 5 Average Arrival Delay	**Metric 6** Controllable Loss (Late Aircraft Factor)	**Metric 7** High-Risk Airline Count	**Metric 8** Mega-Hub Congestion Factor

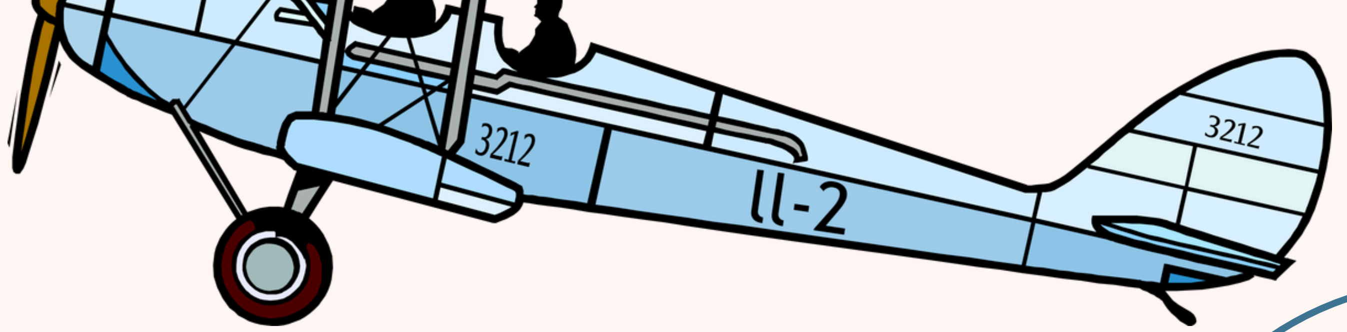
Idea Name	**Layer 0 dimension**	**Layer 1 dimension**	**Layer 2 dimension**	**Is there anything important we missed?**
View 1: Flight Ops Overview	- OTP Rate- Delay Minutes- Total Flights- Cancel Rate	- By Time (Month)- By Airline Strategy	- Time + Airline	Trend lines explicitly tracking OTP and Delay fluctuations over Time
View 2: Delay Root Causes	- Late Aircraft Delays- Weather Delays- Airline Delays- NAS Delays	- By Hour Block- By Route (Origin)	- Hour + Delay Category	Tracking the cascading snowball effect of evening delays.
View 3: Airline & Airport Ops	- Average Arrival Delay- OTP Rate- Flight Volume	- By Hub (Origin/Dest)- By Carrier Status	- Carrier + Hub	Scatter plot identifying exact outlier airlines generating severe delays.
View 4: Temporal Drivers	- Total Delay Minutes- Delay Count	- By Season- By Day of Week	- Day of Week + Season	Day-of-Week peak heatmaps.

STAGE 3: IDEATE – STRUCTURE IDEA

Scorecard	*OTP Rate*	*Total Delay Minutes*	*Cancellation %*	*Flights Volume*	*Average Arrival Delay*

Idea Name	Very Important Information	Important Information	Detailed Information
View 1: Flight Ops Overview	High-level tracking of OTP Rate, Total Delay Minutes, and Flight Volume.	Trend lines showing OTP fluctuations and Total Delay mapped by Month and Carrier.	Top Carriers by Volume. Overall System Delay breakdown.
View 2: Delay Root Causes	Identifying exact outliers ("Which factors drive cascading delays?").	Granular tracking of Delay Categories (Late Aircraft, Weather, NAS) by Time of Day.	Deep dive identifying 16:00-19:00 cascading evening failures to trigger schedule buffer decisions.
View 3: Airline & Airport Ops	Scatter plots mapping On-Time Rate vs Average Arrive Delay for Carriers.	Geographic mapping of congested hubs (e.g., ORD, JFK) causing network pressure.	Precision tracking comparing High-Risk vs Reliable performers to optimize operations.
View 4: Temporal Drivers	Heatmaps of Seasonal (Summer/Winter) and Day-of-Week delay severity.	Highlighting peaks around major Q1 winter storms (Jan 27, Mar 5).	Deep route drill-down evaluating whether bad days are localized storms or systemic collapses.





STAGE 4: PROTOTYPE & REVIEW

Review Area	Quality of Information	Actionable Output
Operations Efficiency Monitoring	Are the System OTP and Total Delays trends immediately clear?	If No: Enhance KPI cards to clearly show raw delayed flight volume vs on-time volume.
Bottleneck Optimization	Can the ops team instantly spot airlines driving severe delays with poor results?	If No: Ensure the scatter plot explicitly separates outliers (low OTP, high delay) from the baseline cluster.
Root-Cause Alignment	Is it easy to distinguish whether delays are driven by Weather or Internal Airline factors?	If No: Redesign the Root Cause panel to explicitly split "Controllable" vs "Uncontrollable" delay minutes.